

ANIMAL BIRTHDAY PREDICTOR (ABP)

MILESTONE 2 PROGRESS & NEXT-STAGE CLIENT REQUIREMENTS REPORT

Project: Animal Birthday Predictor Mobile App	Milestone 2 Start Date: 20 August 2026
Technology Stack: Flutter, Dart, Riverpod, Supabase	Milestone 2 Status: 100% COMPLETED (All 21 Deliverables)

1. Milestone 2 Deliverables & Technical Progress Notes

Milestone Deliverable	Status	Implementation Details & Technical Notes
Pregnancy Module	DONE 	Full equine gestation module supporting Natural, Chilled, Frozen, & ICSI/ET methods with live countdown timers and progress tracking.
Pregnancy Details	DONE 	Detailed view with circular progress bar, trimester breakdown, expected foaling countdown, and days elapsed.
Three Pregnancy Scans	DONE 	Automatic calculation & tracking for Scan 1 (Day 14–16), Scan 2 (Day 30 Heartbeat), and Scan 3 (Day 45 Organogenesis).
Advanced Pregnancy Info	DONE 	Comprehensive biological timeline detailing monthly fetal milestones, maternal changes, and veterinary guidance.
Caslick Information	DONE 	Dedicated Caslick procedure tracking, surgical history logging, and pre-foaling opening reminders.
Fetal Sex	DONE 	Ultrasound fetal gender determination logging (Colt, Filly, Unknown) recorded during veterinarian scan sessions.
Preventative Care	DONE 	Complete wellness tracking system for mares, foals, and puppies with historical log entries and due date alerts.
Vaccination Module	DONE 	Standard equine vaccination schedule (Pneumabort-K at 5, 7, 9 months; Tetanus/Flu booster at 10 months).
Dentist Records	DONE 	Equine dental checkup logging, floating records, practitioner notes, and next scheduled exam tracking.
Farrier Records	DONE 	Hoof care and shoeing/trimming log with farrier contact association and 6–8 week recurring cycle tracking.
Phone Dial Integration	DONE 	AppPhoneLauncher integrated across Vet, Breeder, and Farrier cards with direct native dialer and clipboard fallback.
Record Detail Menu	DONE 	Contextual quick action menus on animal cards providing one-tap navigation to edit, view, scans, and care logs.
Complete Record Detail View	DONE 	Unified profile screen consolidating pedigree, microchip, breeding records, preventative health, and photos.
Edit Existing Records	DONE 	Full multi-attribute editing support for animal profiles, breeding entries, and dates with database synchronization.
Pregnancy Scan Editing	DONE 	In-place editing of scan dates, confirmation status, ultrasound notes, and attending veterinarian details.
Persist Scan Confirmations	DONE 	Permanent database persistence for Scans 1, 2, and 3 confirmation flags across app restarts and state refreshes.
Recipient Photo Saving	DONE 	Dedicated photo capture and cloud/local storage integration specifically for surrogate/recipient mares in ET workflows.
Android Image Rendering Fix	DONE 	Resolved memory handling and aspect ratio rendering issues for camera/gallery image previews on Android devices.
Multiple Images Support	DONE 	Multi-photo management allowing animal profile avatars, ultrasound scan captures, and markings reference pictures.

Complete CRUD Operations	DONE (■)	Create, Read, Update, and Delete operations fully wired across Animals, Pregnancy Records, Care Logs, and Contacts.
Testing of Horse Workflow	DONE (■)	Verified complete equine lifecycle flow: Mare Registration → Breeding Method → Scans 1/2/3 → Foaling Due Date calculation.

2. Client Feedback & Next Stage Feature Analysis (12 Items)

The following table evaluates each item requested in the client's latest feedback, categorized by current implementation state:

#	Client Requirement	Status	Technical Analysis & Next Stage Action
1	45-Day Scan Certificate (AI & Recipient Mare)	MISSING	Day 45 scan data is recorded, but dedicated official 45-Day Scan PDF Certificate generator and UI button must be created.
2	Due Date / Foaling Diary (20–100 Mares & PDF)	MISSING	Centralized stud foaling calendar, paddock movement planner, and batch exportable PDF foaling diary need to be built.
3	Dedicated Due Date Page ('When is my foal due?')	CODE READY / PARTIAL	Calculation engine is 100% functional. Standalone prominent 'When is my foal due?' screen needs to be wired to drawer & dashboard.
4	Twin Warning on Scan 1 (Day 14–16 Alert & Re-scan)	MISSING	Interactive twin detection toggle on Scan 1, prominent alert modal, and urgent re-scan schedule prompt must be added.
5	Calendar + Diary Sync (Multi-module sync)	CODE READY / PARTIAL	Domain entities exist; system-wide date synchronization and device calendar integration need to be connected.
6	Phone Functionality (Native Dialer & Fallback)	IMPLEMENTED	AppPhoneLauncher is fully active with direct mobile dialer, clipboard fallback for non-mobile, and email composer.
7	Congratulations Section (Celebratory Experience)	PARTIAL (NEEDS UPGRADE)	Screen and routing exist. Celebratory visual enhancements (confetti animations, sound/haptics, milestone badge) are required.
8	ABP Branding (Brand Stamping & Seals)	PARTIAL (NEEDS UPGRADE)	Gold theme and logos exist. Consistent app-wide watermarks, header signatures, and certificate seals need enforcement.
9	Payment Details Page (Pricing / Subscriptions)	MISSING	Dedicated Payment Details / In-App Purchases / Subscription page needs to be created.
10	FAQ Page (Client-Managed Q&A;)	MISSING	Searchable, expandable Frequently Asked Questions page needs to be added to navigation.
11	Disclaimer (Global Legal Disclaimer)	PARTIAL (PENDING TEXT)	In-line disclaimers exist. A dedicated Global Legal Disclaimer page/popup is ready to embed client's finalized text.
12	Final QA / Overall Flow (End-to-End Workflow)	PENDING NEW ITEMS	Full verification across Equine → Scans → 45d Cert → Diary → Congratulations will be finalized after new features are built.

3. Summary of Core Implemented Architecture

A. Authentication & Profile Engine

- Supabase Auth with secure session handling, auto-routing, and password reset deep links.
- Email verification screen with automatic poll-and-verify detection.
- Permanent Account Deletion with cascading record cleanup and safety warnings.

B. Multi-Species Registry & Interactive Markings

- 9 supported animal species with filterable directories and search.
- Interactive visual marking selector for face (Star, Stripe, Blaze) and legs (Sock, Stocking, Pastern).
- Integrated camera and gallery image capture for animal profile avatars.

C. Equine Gestation & Care Schedule Engine

- Exact gestation math according to Section 4 ABP rules (Natural 341d, Chilled 341d, Frozen 340d, ICSI/ET 332d).
- Biological dam vs. surrogate recipient mare tracking for embryo transfer programs.
- Preventative care schedules including Pneumabort-K (5, 7, 9m), deworming, and pre-foaling boosters.

D. Offspring Modules & Official PDF Certificates

- Foal registration with pedigree linking to dam, sire, and recipient mare.
- Canine module with dedicated daily/weekly Puppy Weight Growth Tracker.
- Vector PDF birth certificate generator for Foals and Puppies with direct printing and sharing.

4. Next Stage Implementation Plan

Phase 1: Milestone 3 Priority Features

- Implement **45-Day Scan Certificate Generator** (PDF export & in-app viewer).
- Add **Interactive Twin Warning & Urgent Re-scan Workflow** to Scan 1.
- Create **Dedicated 'When is my Foal Due?' Page** accessible from Home/Drawer.

Phase 2: Stud Diary & Exportable Schedules

- Build **Stud Foaling Diary Matrix** supporting 20–100 mares with paddock movement tags.
- Implement **Exportable Foaling Schedule PDF** for stud barn staff.

Phase 3: Visual Polish & Essential Pages

- Upgrade **Congratulations Screen** with celebratory particle animations & sound/haptics.
- Enforce comprehensive **ABP Brand Watermarks & Seals** throughout screens and exports.
- Add **Payment Details Page**, **Interactive FAQ Page**, and **Global Legal Disclaimer**.

Phase 4: QA & Device Testing

- Perform comprehensive multi-viewport testing across Android, iOS, and tablets.
- Validate dialer actions, PDF generation performance, and state persistence.